# INFORMATION SECURITY & IT LAWS

## IMPORTANT QUESTIONS — ALL 5 UNITS

★ = High-priority question | Page 1: Units 1–3 | Page 2: Units 4–5

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## UNIT 1: INTRODUCTION TO INFORMATION SECURITY

★ What is Information Security? Explain the principles of Information Security.

- Explain the various aspects of Information Security (PAIN).
- Explain OS security features: authentication, logs, audit features, file-system protection and user privileges.
- Write short notes on Anti-Virus Software and RAID options.
- Write a short note on Intruders and Insiders.

★ Why is operational security important in an organization?

★ What are the various password attacks? How can a system administrator protect against them?

## UNIT 2: NETWORK SECURITY

★ Define intrusion. How does a firewall help protect against it?

★ What are the benefits and services provided by IP Security (IPSec)?

★ Explain the Network Security Model with a suitable diagram.

- Explain security weaknesses in IP, TCP, UDP, RIP, OSPF, HTTP and SMTP protocols.
- Explain security weaknesses of hubs, switches, routers and Wi-Fi.
- Explain IPSec, HTTPS, VLAN, VPN and ingress filtering as network security solutions.
- Write short notes on VPN and spam.
- Explain the difference between the Internet and Intranet.

## UNIT 3: CRYPTOGRAPHY, PKI AND SOFTWARE SECURITY

★ What is the goal of cryptography? Define symmetric and asymmetric cryptography.

★ Explain the DES algorithm.

★ How is encryption different from hashing? Explain with a diagram.

- Explain the difference between symmetric and asymmetric cryptography.
- Explain Caesar's cipher with an example.
- Explain the Diffie–Hellman key exchange algorithm.
- Write short notes on Digital Signature and Digital Certificate.
- Explain Public Key Infrastructure (PKI).
- Explain secure-code techniques and buffer overflow.
- Explain application hardening and application patches.

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## UNIT 4: NETWORK SECURITY PRODUCTS

★ What is an IDS? Explain its types.

★ Explain a firewall and its working.

• Explain the benefits and working of a VPN.

• What is a VPN concentrator? Explain its functions.

• Explain content-screening gateways and their importance.

• How do firewalls and IDS/IPS help protect a network?

## UNIT 5: SECURITY STANDARDS, IT LAWS AND BUSINESS CONTINUITY

★ Explain Intellectual Property Rights (IPR).

★ Explain Copyright Law and Patent Law.

★ What is the Information Technology Act, 2000? Explain its important provisions.

★ Explain the process of authenticating electronic records under the IT Act, 2000.

★ What are the different Information Security Standards? Explain ISO 27001.

• Explain security-audit procedures and their importance.

• Explain the development and importance of security policies.

★ Explain Disaster Recovery Planning (DRP) and Business Continuity Planning (BCP).

***Study note:*** *This is a syllabus-matched priority list, not a guarantee of exam questions. Prepare definitions, diagrams, comparisons, working steps and examples where relevant.*

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